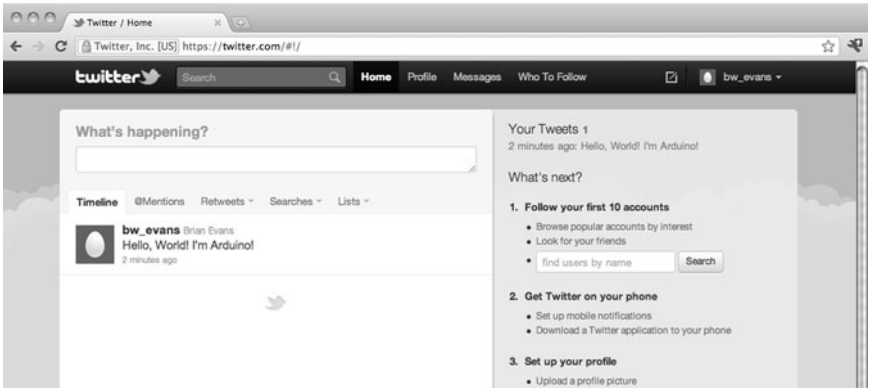


Figure 11-3. *Something tweeted*

Okay, so *"Hello, World! I'm Arduino!"* is perhaps not the most exciting thing to tweet, but it's a start right? With the right kind of sensors, I'm sure you can find something interesting to make tweet using these libraries. Or maybe instead, you can get the Arduino to display incoming tweets on an LCD or other form of display. For more ideas on what to do with the Arduino Ethernet Shield, check out the following tutorials from bildr, the modular tutorial site:

```
http://bildr.org/2011/06/arduino-ethernet-pin-control/
http://bildr.org/2011/06/arduino-ethernet-client/
```

Bonus Project 2: Make Something Move

We also didn't get a chance to talk much about making things move. We did have a look at connecting a small DC fan back in Chapter 6, and in Chapter 9 we discussed servo and stepper motors as way to make something move with fairly precise positioning. You might instead want to make something move using common and very popular DC motors. As explained in more depth in the next chapter, DC motors come with and without gearboxes and run at a variety of speeds with different current requirements. The following are three possible things that we can look for when connecting DC motors to the Arduino:

- The motor can run at variable speeds
- The motor can run at high currents
- The motor can switch directions

Of these three things—in a simple circuit that we are not throwing the kitchen sink at—we can usually do two of these at any one time. So for example, we can run a motor at high currents and at a range of speeds, but it's a little more difficult to get it to switch directions. Instead, we could have a motor change direction, but we would then need to decide whether a high current or variable speeds is more important.

The code to work with DC motors is fairly simple using a combination of digital and/or analog outputs, which we covered somewhat thoroughly in Chapters 5 and 6. The circuits, however, are a little more complex, and we won't be able to go into all the complexities here, but let's cover three simple ways to interface DC motors that illustrate the rule of variable speeds/high currents/switching directions